**MECHANICAL PROPERTIES**

| Product Denomination | GG200T(M40J)-DT120-42 | | | |
|---|---|---|---|---|
| Weaving Style | TWILL | | | |
| Yarn Type | TORAY M40J | | | |
| F.A.W. ($g/m^2$) | 200 | | | |
| Nominal Fiber Volume Fraction | 49% | | | |
| Cure Conditions vs. Mechanical Data | Autoclave cure: Cured 90 min @ 120°C / 6 bar | | | |
| Average Density of Cured Sample ($g/cm^3$) | 1,49 | | | |
| Nominal CPT (mm) | 0,23 | | | |
| **Test Conditions** | Room Temp. | | | |
| TEST ID | CM14-41 | | | |
| **TENSILE 0°** | ASTM D3039 | | | |
| Actual CPT (mm) | 0,23 | | | |
| Tensile strength (MPa) | 864 | | | |
| Tensile modulus (GPa) | 86,8 | | | |
| Calculated Ultimate Elongation (%) | 1,00 | | | |
| Tensile Poisson ratio (-) | 0,06 | | | |
| **TENSILE 90°** | ASTM D3039 | | | |
| Actual CPT (mm) | 0,23 | | | |
| Tensile strength (MPa) | 817 | | | |
| Tensile modulus (GPa) | 84,2 | | | |
| Calculated Ultimate Elongation (%) | 0,97 | | | |
| **COMPRESSION 0°** | ASTM D6641 | | | |
| Actual CPT (mm) | 0,23 | | | |
| Compression strength (MPa) | 496 | | | |
| Compression modulus (GPa) | 77,7 | | | |
| **COMPRESSION 90°** | ASTM D6641 | | | |
| Actual CPT (mm) | 0,22 | | | |
| Compression strength (MPa) | 483 | | | |
| Compression modulus (GPa) | 76,8 | | | |
| **IN-PLANESHEAR** | ASTM D3518 | | | |
| Shear strength of 5% Strain (MPa) | - | | | |
| Maximun Shear strength (MPa) | 88,2 | | | |
| Shear modulus (GPa) | 3,04 | | | |
| **ILSS** | ASTM D2344 | | | |
| ILSS (MPa) | 62,20 | | | |

All data are normalised to Nominal fibre volume fraction (except for ILSS and In-Plane Shear).

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Checked By Nicola Del Debbio 23/08/2021